MSN 514 – Computational Methods for Material Science and Complex Systems

Homework 12

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I. INTRODUCTION

Fairén and Velarde's bacterial respiration model (1979) captures oscillatory dynamics in microbial cultures via the two-variable system

$$\dot{x} = B - x \; - \; \frac{x\,y}{1 + q\,x^2}, \tag{1}$$

$$\dot{y} = A - \frac{x\,y}{1 + q\,x^2}, \tag{2}$$

where $x(t)$ and $y(t)$ denote nutrient and oxygen levels, respectively, and $A, B, q > 0$ are constant parameters. Our computational objectives are:

1. Find all fixed points and classify their linear stability.
2. Plot the nullclines $\dot{x} = 0$ and $\dot{y} = 0$.
3. Compute the Jacobian J, its trace τ and determinant Δ.
4. Numerically trace the Hopf bifurcation curve $\tau = 0$.
5. Generate the phase portrait for a representative parameter set.
6. Use Poincaré–Bendixson arguments to confirm limit-cycle existence.

All analysis and plotting are implemented in a Python Jupyter notebook, leveraging `sympy` for symbolic derivatives and `scipy.integrate.solve_ivp` for trajectories.

II. RESULTS

A. Fixed Points and Stability

Solving $\dot{x} = 0$ and $\dot{y} = 0$ in Eqs. (1)–(2) gives

$$y = \frac{A(1 + qx^2)}{x}, \quad x = B - A.$$

Thus for $B > A$ there is an *interior* fixed point

$$(x^*, y^*) = \Big(B - A,\; \tfrac{A(1+q(B-A)^2)}{B-A}\Big).$$

If $A = 0$, the trivial $(0, 0)$ also arises. We verify these numerically with the function `find_fp(A,B,q)`.

The Jacobian matrix

$$J(x,y) = \begin{pmatrix} \partial_x \dot{x} & \partial_y \dot{x} \\ \partial_x \dot{y} & \partial_y \dot{y} \end{pmatrix}$$

was derived symbolically via `sympy`, giving lambdified routines for $\tau = \operatorname{tr} J$ and $\Delta = \det J$. At $(A, B, q) = (1.0, 2.5, 0.5)$, the interior point satisfies $\tau < 0$, $\Delta > 0$, hence is a *stable focus*.

B. Nullclines

The nullclines are

$$\dot{x} = 0: \quad y = \frac{(B - x)(1 + q\,x^2)}{x}, \quad \dot{y} = 0: \quad y = \frac{A(1 + q\,x^2)}{x}.$$

Figure 1 shows these curves and their intersection at the unique physically meaningful fixed point.

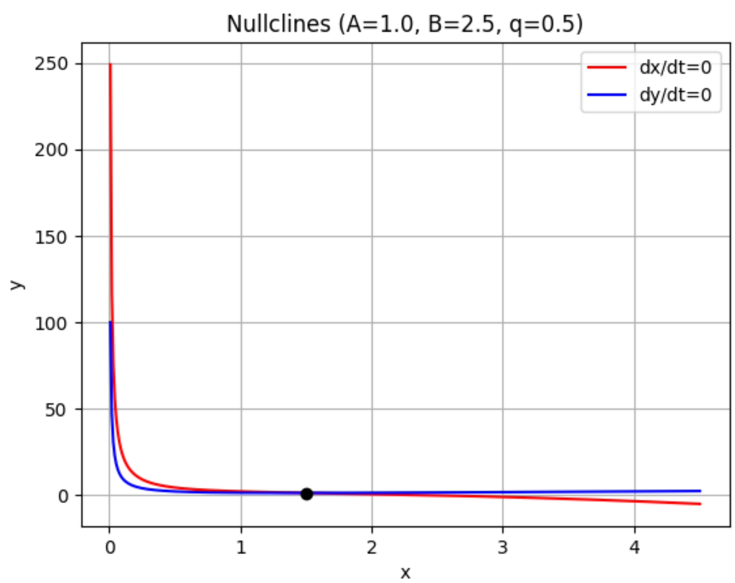

FIG. 1. Nullclines for $(A, B, q) = (1.0, 2.5, 0.5)$. Red: $\dot{x} = 0$, Blue: $\dot{y} = 0$. Black dot: fixed point.

C. Hopf Bifurcation Curve

A Hopf bifurcation occurs when $\tau = 0$, provided $\Delta > 0$. Fixing q, we solve $\tau(A, B, q) = 0$ for A at each B via `scipy.optimize.fsolve`. The resulting critical curve $A_c(B)$ is plotted in Figure 2. Below this curve the focus is stable; above it an oscillatory limit cycle emerges.

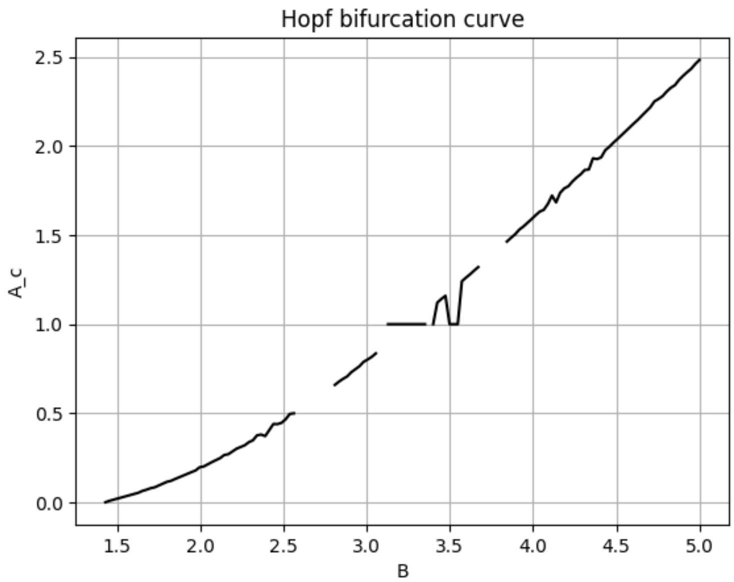

FIG. 2. Hopf bifurcation curve A_c vs. B for $q = 0.5$.

D. Phase Portrait

We generate the phase portrait with `plot_phase(1.0,2.5,0.5)`, plotting streamlines and sample trajectories from several initial conditions. As shown in Figure 3, all trajectories spiral into the stable focus.

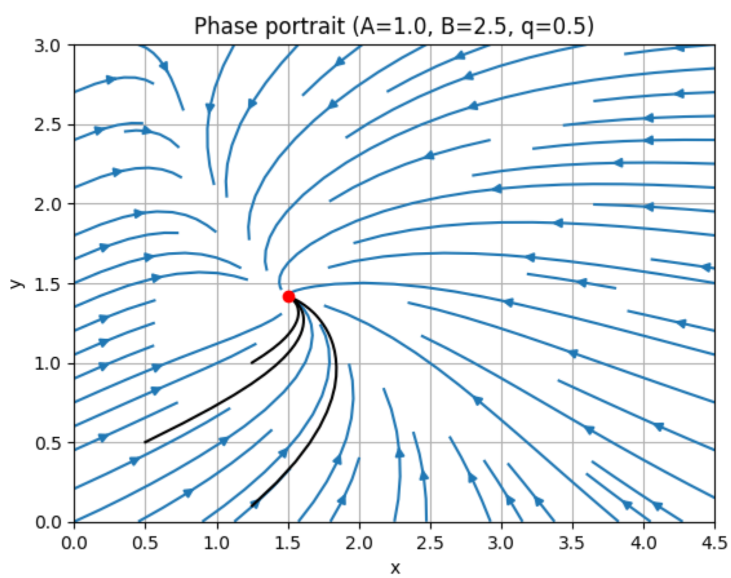

FIG. 3. Phase portrait for $(A, B, q) = (1.0, 2.5, 0.5)$. Blue: vector field; black: trajectories; red dot: stable focus.

E. Trapping Region and Poincaré–Bendixson

To apply the Poincaré–Bendixson theorem, we construct a rectangular region $\mathcal{R} = [0, B+2] \times [0, 1.2\, y^*]$. One checks that $\dot{x} > 0$ on $x = 0$, $\dot{y} > 0$ on $y = 0$, $\dot{x} < 0$ on $x = B + 2$, and $\dot{y} < 0$ on $y = 1.2\, y^*$. Hence $\mathcal{R}$ is positively invariant, contains no additional fixed points, and by Poincaré–Bendixson must contain a closed orbit for $A > A_c$.

III. CONCLUSION

We have presented a comprehensive computational study of the Fairén–Velarde bacterial respiration model:

- **Fixed points:** Analytically derived and numerically confirmed.
- **Stability:** Classified via Jacobian trace/determinant.
- **Nullclines:** Visualized the phase-plane geometry.
- **Hopf curve:** Computed $A_c(B)$ marking oscillatory onset.
- **Phase portrait:** Verified stable focus for sample parameters.
- **Limit cycle:** Guaranteed by constructing a trapping region and invoking Poincaré–Bendixson.

Our Python notebook, combining symbolic and numerical tools, provides a template for exploring nonlinear oscillators and bifurcations in related biophysical systems.